

Course 6

# Ethics in Research

# Why?



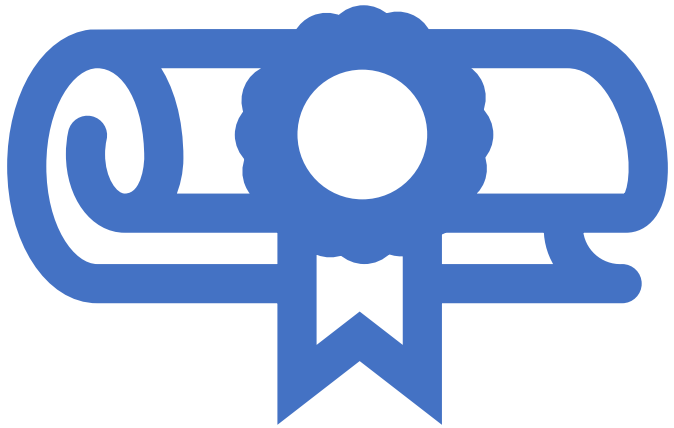






Academics

Bachelor Thesis



# Progress of our faculty

- Turnitin: plagiarism detection service
- **2015(ᵔ)**: PhD & Habil. Theses
- **2020**: introduced to dissertation theses (master)
- **2021**: all theses: Bachelor and Master graduates

# Copyright Law

- ✓Art 33(1)b: “utilizarea de scurte citate dintr-o operă, în scop de analiză, comentariu sau critică ori cu titlu de exemplificare, în măsura în care folosirea lor justifică întinderea citatului”

  *“use of short passages… for analysis, comment, … example…”*

- ✓Art 33(2)b: .. sunt permise reproducerea, distribuirea, radiodifuzarea sau comunicarea către public, fără un avantaj direct sau indirect, comercial sau economic: de scurte fragmente ale conferinţelor, alocuţiunilor, pledoariilor şi ale altor opere de acelaşi fel, care au fost exprimate oral în public, cu condiţia ca aceste utilizări să aibă ca unic scop informarea privind actualitatea;”

- **Art. 48.** - **(1) ”**Prin contractul de editare, titularul dreptului de autor cedează editorului, în schimbul unei remuneraţii, dreptul de a reproduce şi de a distribui opera”

  *”… the holder of the copyright transfer the right to the editor … in some conditions”*

# Copyright Law - infrigements

- **Art. 1398. -** Constituie infracţiune şi se pedepseşte cu închisoare de la 6 luni la 3 ani sau cu amendă punerea la dispoziţia publicului, inclusiv prin internet ori prin alte reţele de calculatoare, fără drept, a operelor sau a produselor purtătoare de drepturi conexe ori de drepturi sui-generis ale fabricanţilor de baze de date sau a copiilor acestora, indiferent de suport, astfel încât publicul să le poată accesa în orice loc sau în orice moment ales în mod individual.

  *Infrigement – jail 6 month2 to 3 years – expose to public copyright owned items*

- **Art. 1399. -** Constituie infracţiune şi se pedepseşte cu închisoare de la 6 luni la 3 ani ori cu amendă reproducerea neautorizată pe sisteme de calcul a programelor pentru calculator în oricare dintre următoarele modalităţi: instalare, stocare, rulare sau executare, afişare ori transmitere în reţea internă.

  *Infrigement – jail 6 month2 to 3 years – unauthorized reproduction on computing systems*

- **Art. 140. - (1)** Constituie infracţiuni şi se pedepsesc cu închisoare de la o lună la un an sau cu amendă următoarele fapte comise fără autorizarea sau consimţământul titularului drepturilor recunoscute de prezenta lege:

- **a)** reproducerea operelor sau a produselor purtătoare de drepturi conexe

  *Infrigement – jail 1 month to 1 year – reproduce copyright owned items*

# Reference

- Idea, concept, definition, methodology etc.
- Also: figures, tables
- Important! – all information to retrieve the bibliographical source
- Standards (AMS, IEEE)
- Computer Science:
  - References in alphabetical order
  - Dedicated section : *References*

    (not footnotes)
- See: https://www.cs.ubbcluj.ro/wp-content/uploads/ghid-finalizare-2021.docx

# See **** and ***

# Case studies

# Case study 1

John is a young researcher at University A. He has a briliant idea, that if confirmed by experimental data will be a major breakthrough in his career.

Mary, his girlfriend, works in the same research group, and her expertise includes working with experimental data and validation of theoretical results.

John shares the idea with Mary and they obtain promising results

John discusses the idea with his scientific coordinator, Tom and he recommends to publish the study as a unique author in order to obtain a major advantage

John tells Mary, she agrees and she stops working on the project

? Has John proceeded in an ethical way?

? Why being an author is important in research?

? What is a research contribution?

# Remarks

- Authors rights are not restricted to a paper/idea

- Includes:
  - Dataset
  - Software

Original contributions? Relevant scientific contributions?

- Study (synthesis, summary, comparative)
- Dataset
- Software

# Case study 2

- Prof. **X** is the head of a research group and states the publication rules for everyone from the group:
  - Only those that made a substantial intellectual contribution
  - If any disagreement appears, he solves them
  - He always appear as last author of an article which is the research result of the laboratory

# Case study 2

- **A** is a postdoc researcher working on a project
- **B** is a technician, working with **A**, in charge of experimental setup
- **C** is a Master student, working on the final phase of the project
- ?Publishing order:
  - **A** - first author
  - **B** - second author
  - **C** – should not be author (no substantial intellectual contribution), only in "Acknowledgement" part of the paper
  - **X** - last author, as head of research group, and the project was funded by the research group

# Recommended:

- Elsevier: For transparency, we encourage authors to submit an author statement file outlining their individual contributions to the paper using the relevant CRediT roles: Conceptualization; Data curation; Formal analysis; Funding acquisition; Investigation; Methodology; Project administration; Resources; Software; Supervision; Validation; Visualization; Roles/Writing - original draft; Writing - review & editing. Authorship statements should be formatted with the names of authors first and CRediT role(s) following

All authors contributed equally to this work.

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# Case study 3

- Professor **T** is editor in chief of a journal
- He receive a submission from prof. **X**, in a certain domain
- His task is to send for review the paper to prof. **Y**, specialist in that domain and working in the same university as prof. **X**
- **Scenario 1**: prof. **Y** rejects the paper as ”without proper scientific proof” and after 3 months sends a similar paper to the same journal

? Any ethical issues related to **T** and **Y** behavior?

- **<u>Scenario 2</u>**: prof. **Y** accepts the paper of prof. **X**, it is published, and after 3 months prof. **T** receives an email stating that the published results are false.

? What must prof. **T** do?

# Remarks

- Reviewing is an important activity
- Confidentiality
- Respect ethical principles
  - Conflict of interests

Reviewers of articles, conference submissions need to declare conflict of interests, such as: work in same department/projects, published papers together, family relations

- https://link.springer.com/article/10.1007/s10664-020-09821-y

- The authors have retracted this article Palomba and Zaidman (2019). Upon re-review of the experiment presented in the article, the authors identified errors in the flaky test detection strategy. After careful analysis of the replication study, the results presented in this article are rendered unreliable. All authors agree to this retraction.

# Case study 4

- Prof. D is leading a research project regarding how different perspective of the source code affect code comprehensibility and uses a tool
- Need of an empirical study to confirm/infirm tool usability
- Ask the students to use the proposed tool and others to collect data

? “Threats to validity”?

? Are the study results relevant?

# Remarks

- Empirical studies (experiments, case studies, interviews, surveys) must respect standards
- Studies involving persons – special attention

# Case study 5

- Dr. **P** works in a research center at a project regarding improvement of software processes
- Collects data from company **Q** which works at a complex software system for client **R** with a large development team;
- Using the proposed methodology, identifies 5 big defects related to processes and publishes the results in a paper

? Ethical issues?

? Who must be informed and when?

# Remarks

- Studies in collaboration with industrial partners must be carry out based on a specific protocol and with continuous updates on all sides

# ACM reported ethical issues

- **ICIMTech2021**
- On September 9, 2021, IEEE contacted ACM with a claim that the Sixth International Conference on Information Management and Technology (ICIMTech2021), published by ACM in the ICPS, was a fake event, and that the purported conference had used the name and front matter of the real conference published by IEEE under the same name.
- Upon investigation, it was determined that the Program Chair listed in the ICPS application had used a pseudonym. This person failed to answer any of the questions put to them during the investigation, and outreach to the authors of the papers published in ICPS also elicited no response.
- Subsequently, ACM's Ethics & Plagiarism Committee concluded that the combination of fraudulent front matter, lack of response from the Program Chair, and lack of response from the authors cast serious doubt on the integrity of the conference, and thus the authenticity of the proceedings.
- The E&P Committee made a formal recommendation that the entire proceedings be retracted, and a motion to this effect was carried at the Publications Board meeting on December 16, 2021. A retraction notice was posted on the conference home page on the ACM Digital Library.

# ACM reported ethical issues

- **IHIP2018**
- On August 12, 2021, the ACM Ethics and Plagiarism Committee received a claim that a paper published by ACM in the Proceedings of the 2018 International Conference on Information Hiding and Image Processing (IHIP2018) was computer-generated.
- The E&P Committee investigated and found that the paper was scientifically meaningless and was likely to have been generated by a computer. The listed "author" did not respond to ACM's inquiries, and it was discovered that three other papers ostensibly written by the same person had recently been removed from two different publishers' websites for similar reasons. A request to the Program Committee for the original reviews of the paper elicited what were deemed to be counterfeited review materials.
- Given the clear evidence of fraud, as well as doubts arising from ACM's communications with the conference organizers, the E&P Committee made a formal recommendation that the entire proceedings be retracted, and a motion to this effect was carried at the Publications Board meeting on December 16, 2021. A retraction notice was posted on the conference home page on the ACM Digital Library.